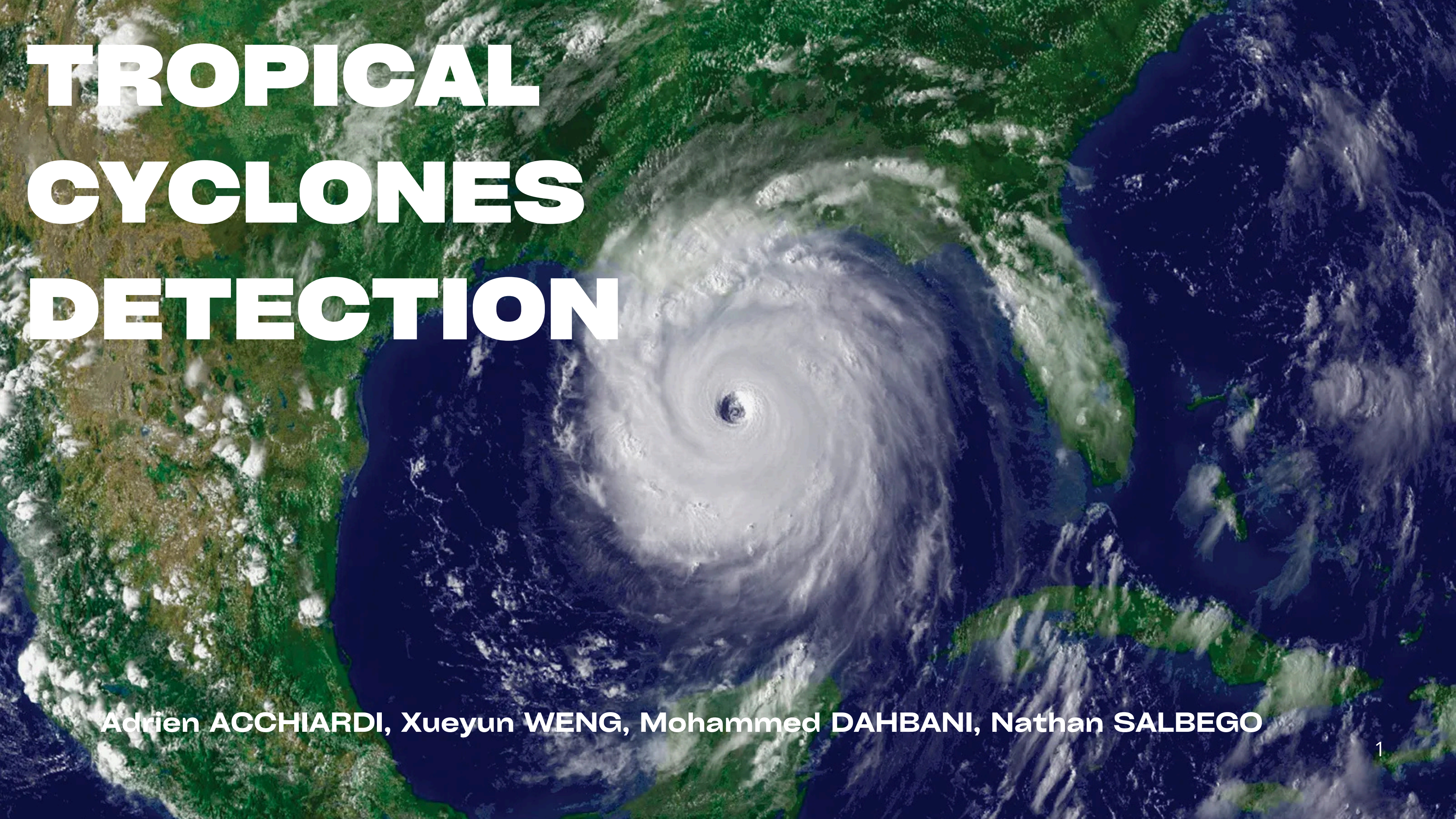




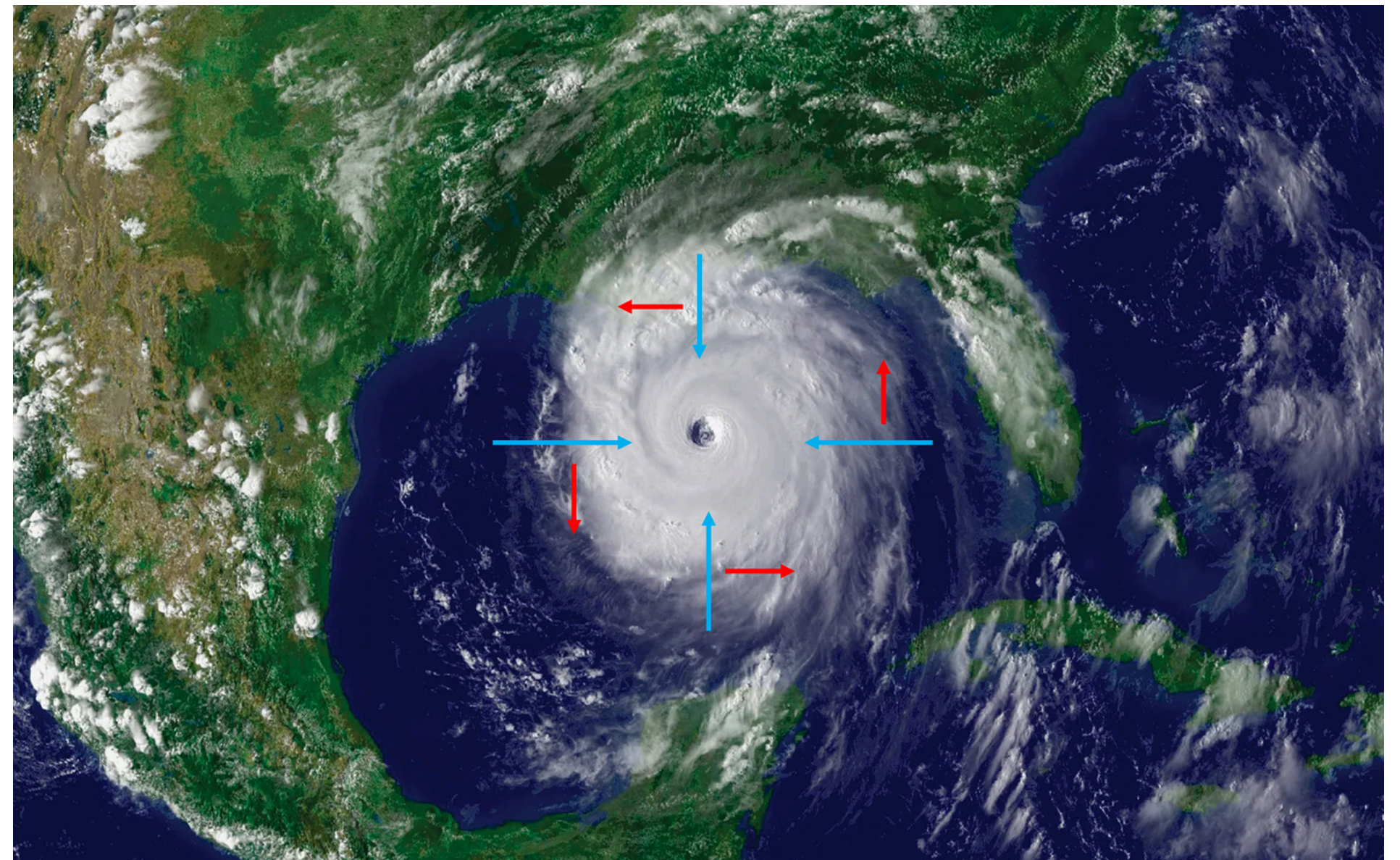
TROPICAL CYCLONES DETECTION

Adrien ACCHIARDI, Xueyun WENG, Mohammed DAHBANI, Nathan SALBEGO

Context

HOW DO TROPICAL CYCLONES FORM ?

- TEMPERATURE AND HUMIDITY
(CONVECTION)
- CORIOLIS FORCE (SPIN)
- PRESSURE GRADIENT (WIND)



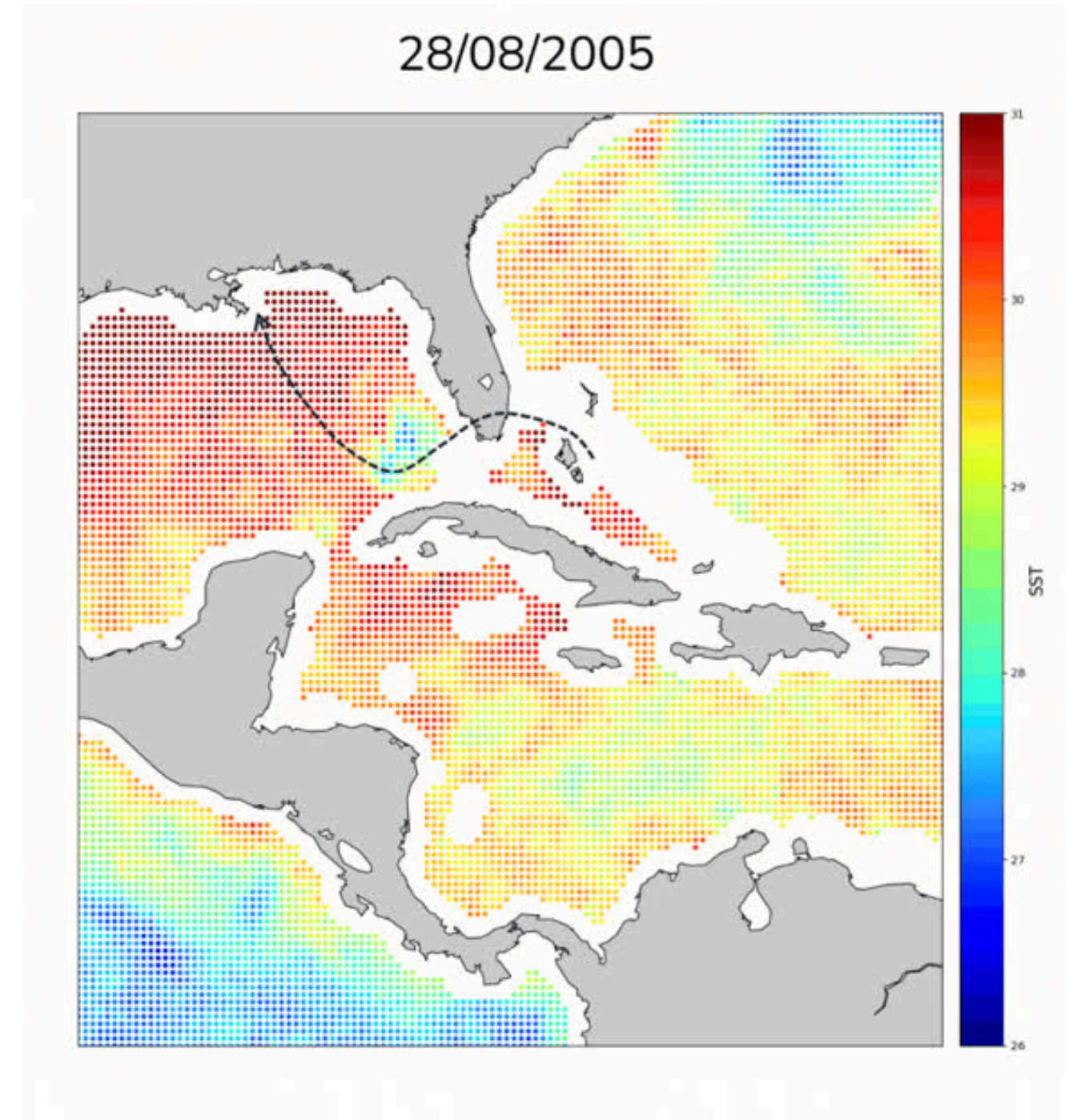
Context

HOW CAN WE OBSERVE A TROPICAL CYCLONE ?

SST (sea surface temperature) can be a relevant indicator



The passage of a hurricane induces a **drop in SST**



HURRICANE KATRINA (2005)

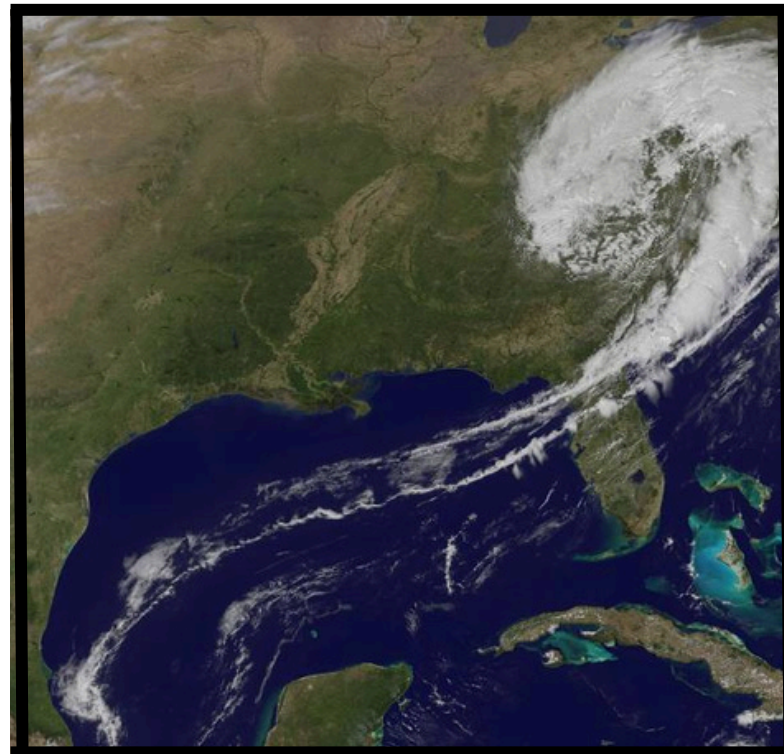
Data and Study zone

The **Dataset** used for our project

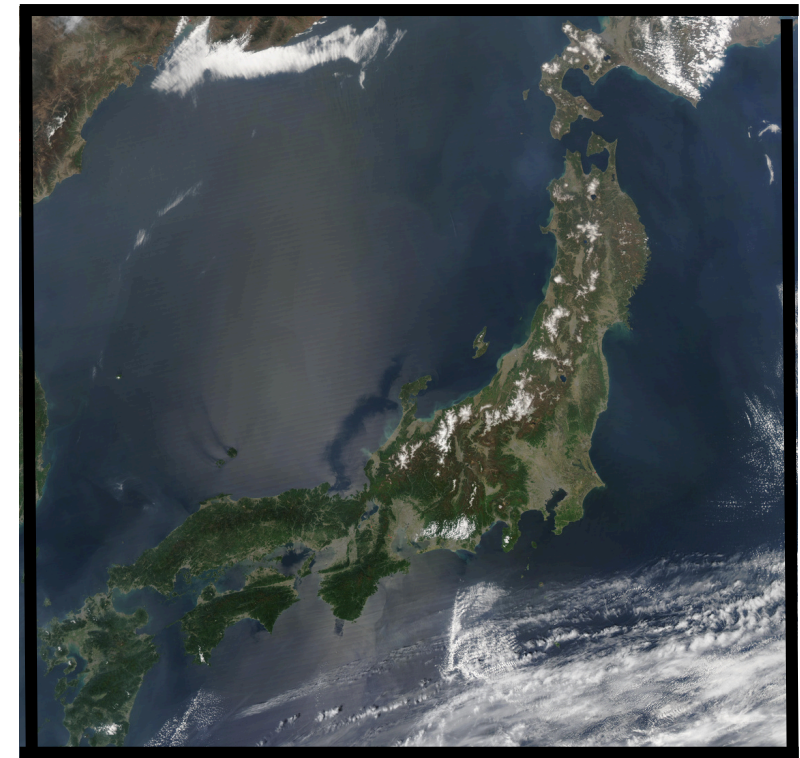
SST satellite data:

- 25 km spatial resolution
- Data from 1998 to 2015
- 1 day time resolution

Focus on two **tropical zones**



Gulf of Mexico



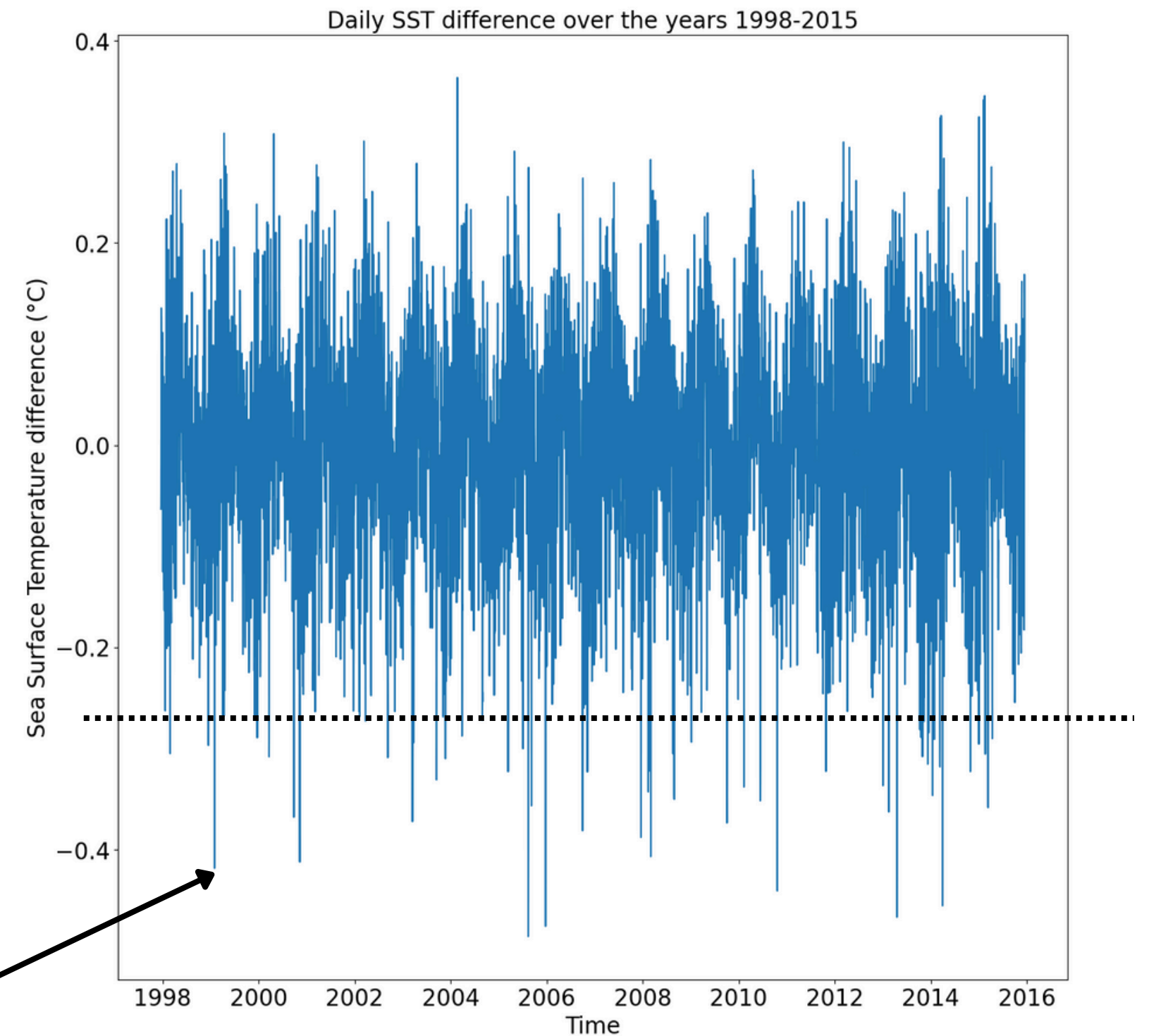
Japan

First approach : Over the entire zone

The easiest approach consists in averaging the SST over the entire area

We can then identify cyclones:

- Average **SST difference between consecutive days**
- We use a **threshold**



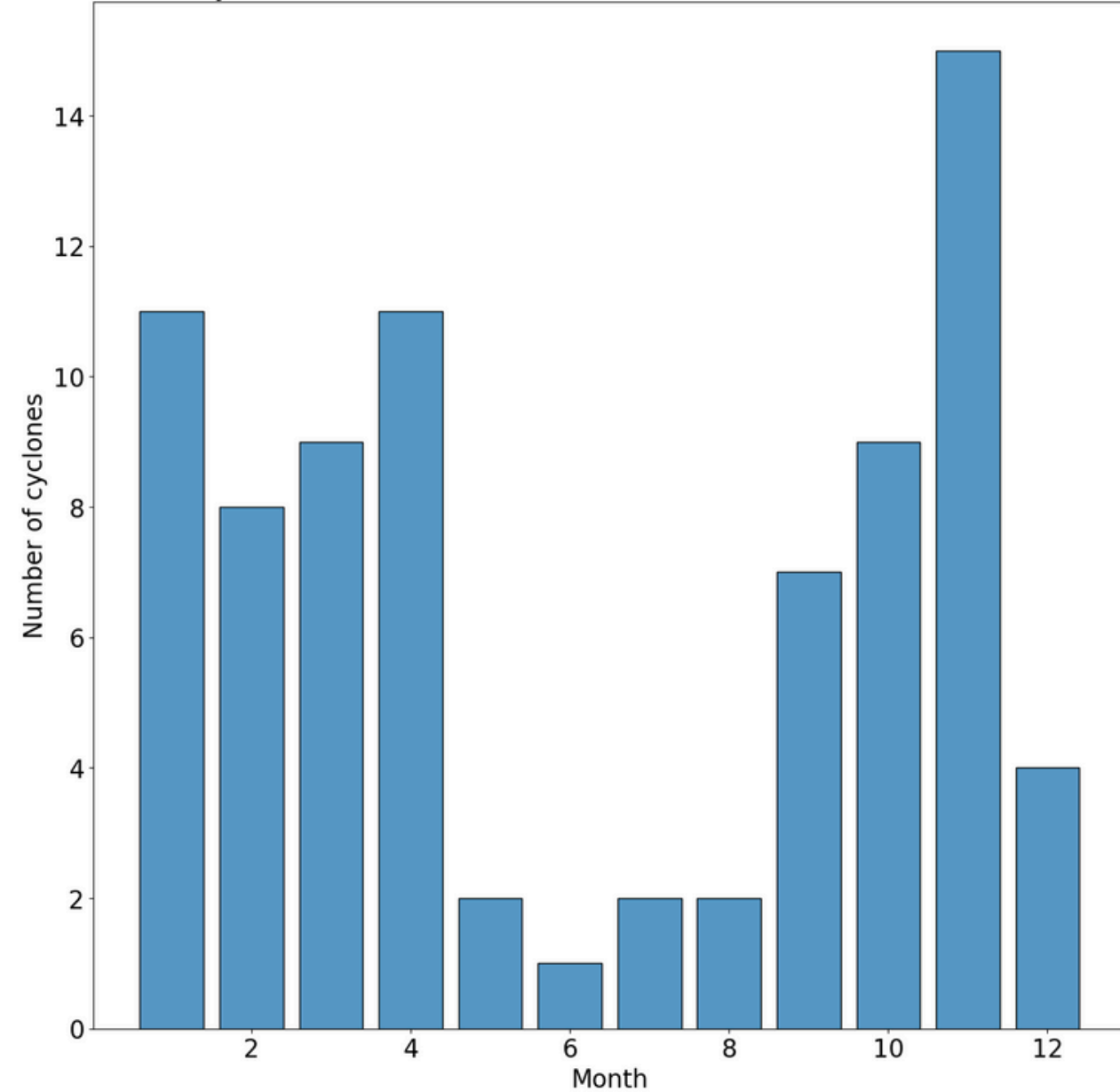
Cyclone

TIME SERIES IN THE GULF OF MEXICO

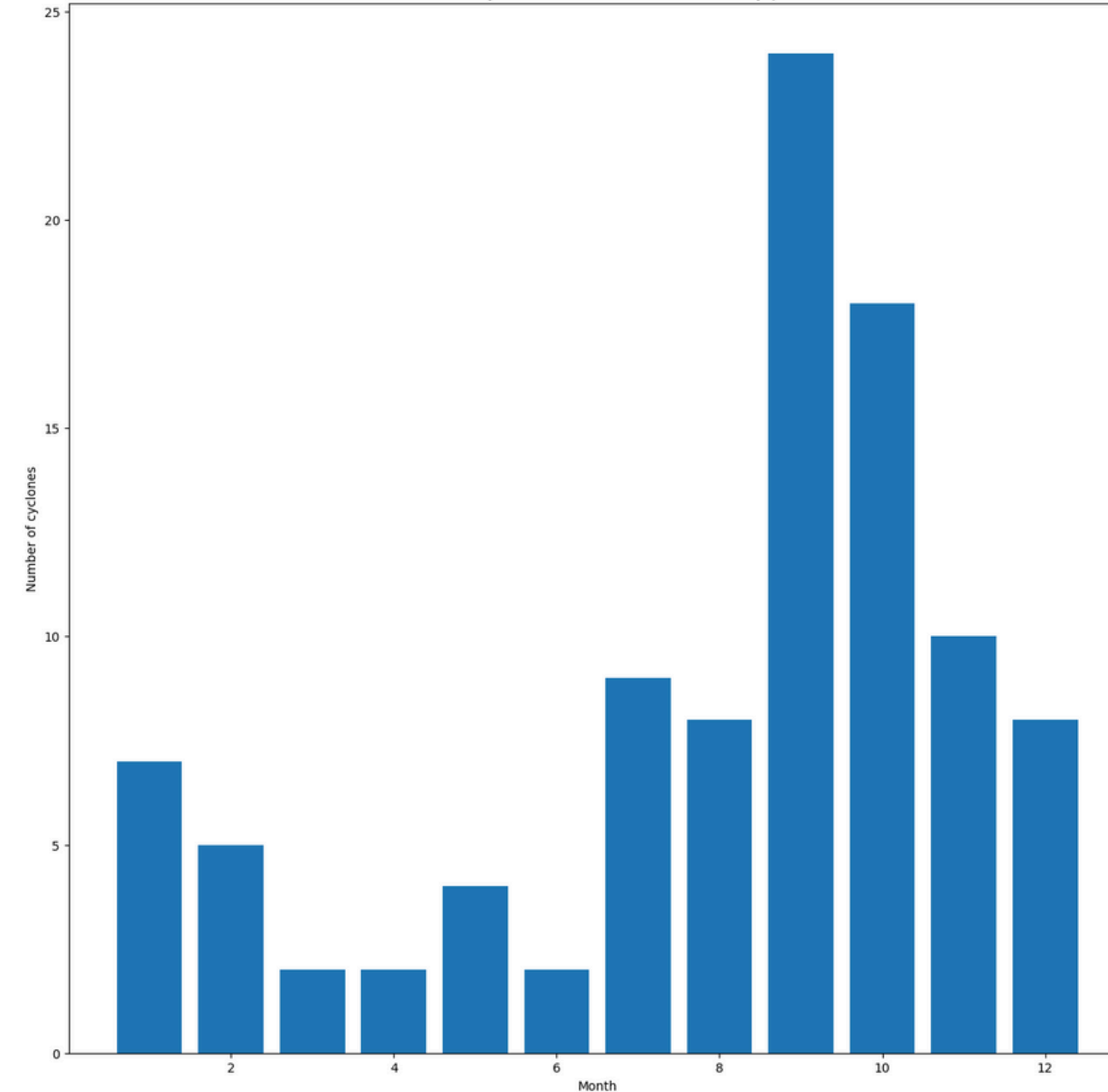
First approach : Over the entire zone

We can now see how often cyclones occur in our studied regions

Number of cyclones for each month between 1998-2015 in the Gulf of Mexico



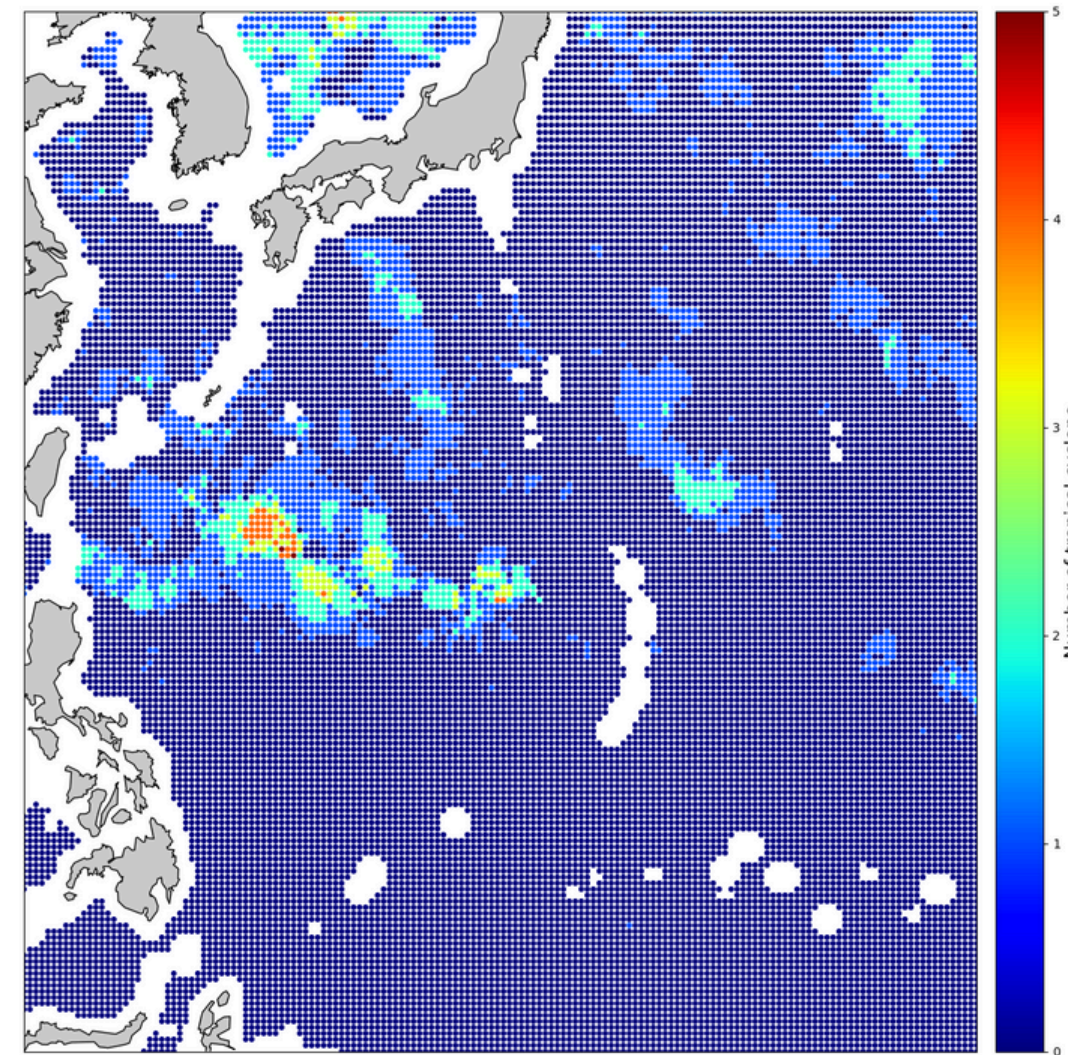
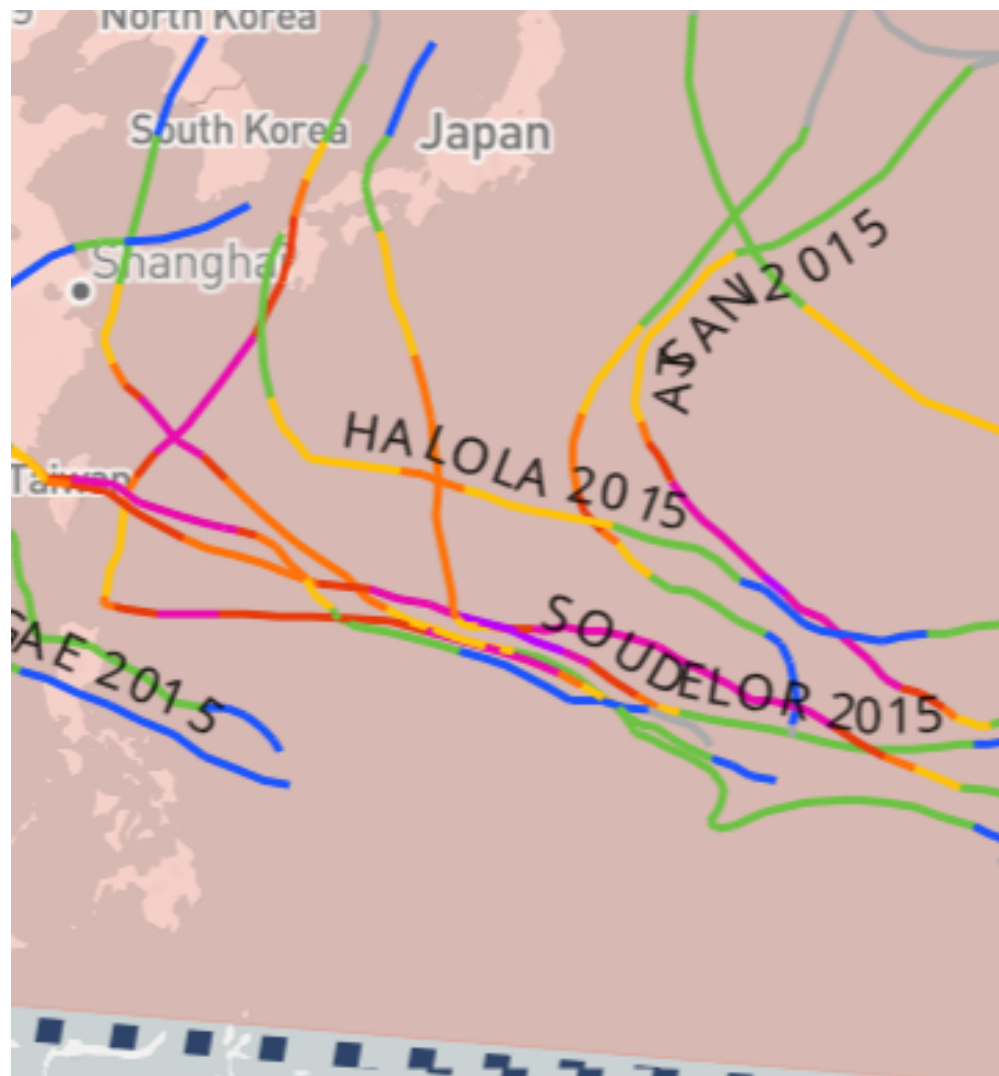
Number of cyclones between 1998 and 2015 in Japan



Cyclones should occur in **summer month**...

Second approach : Point by point

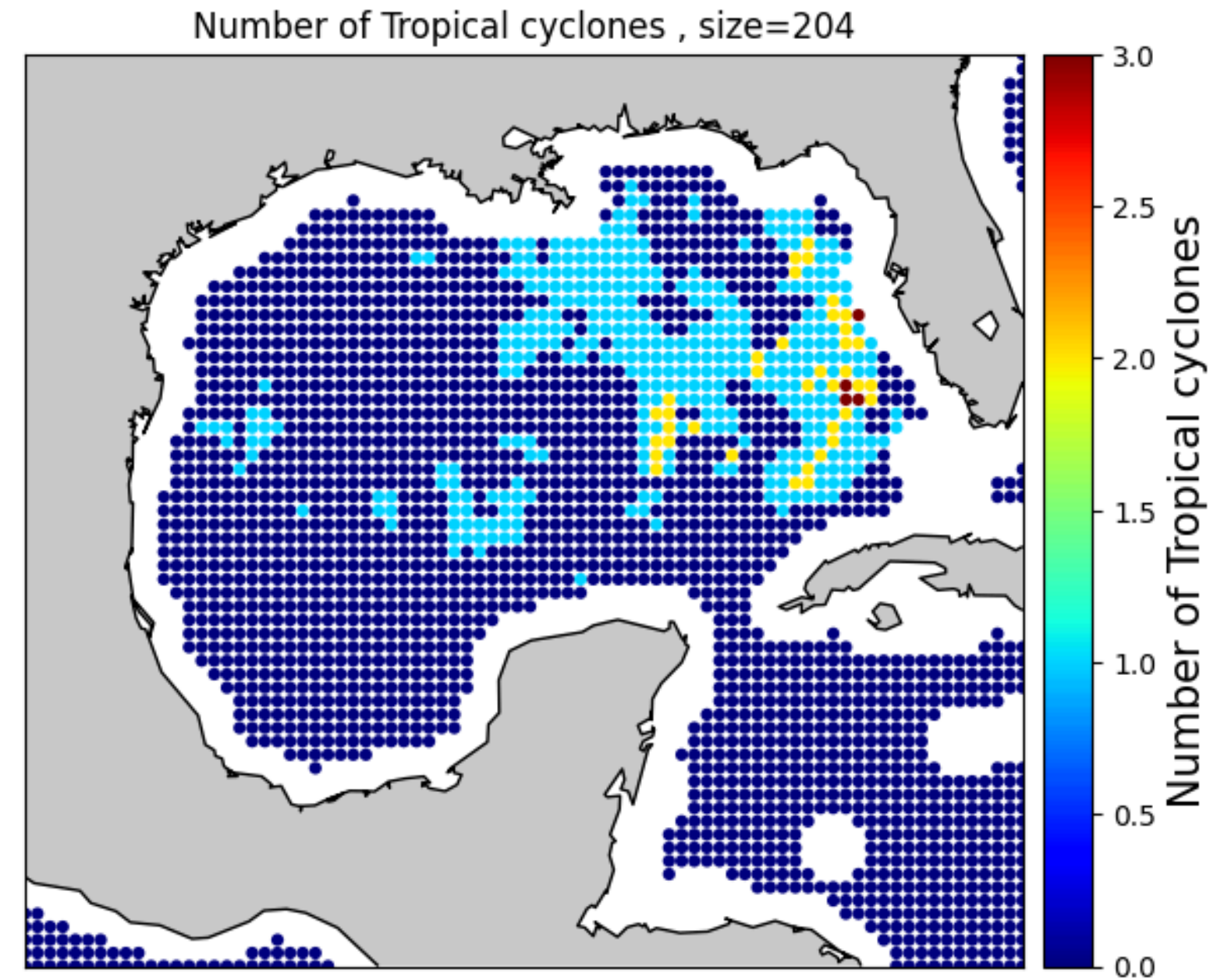
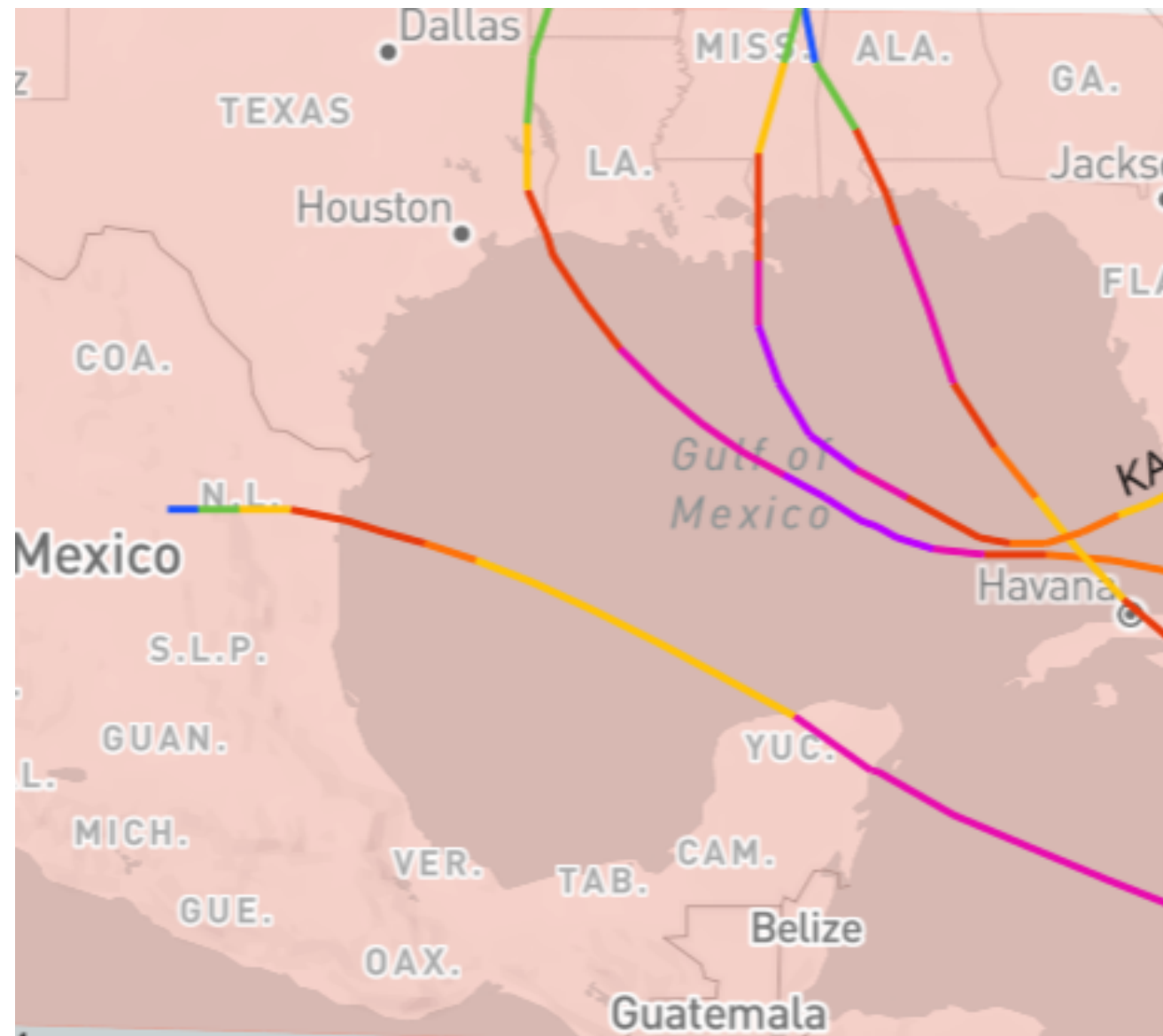
Japan, 2015, June, July, August, September



A more accurate way is to observe locally, given a longitude and a latitude, the SST difference

Second approach : Point by point

Mexico, 2005, June, July, August, September



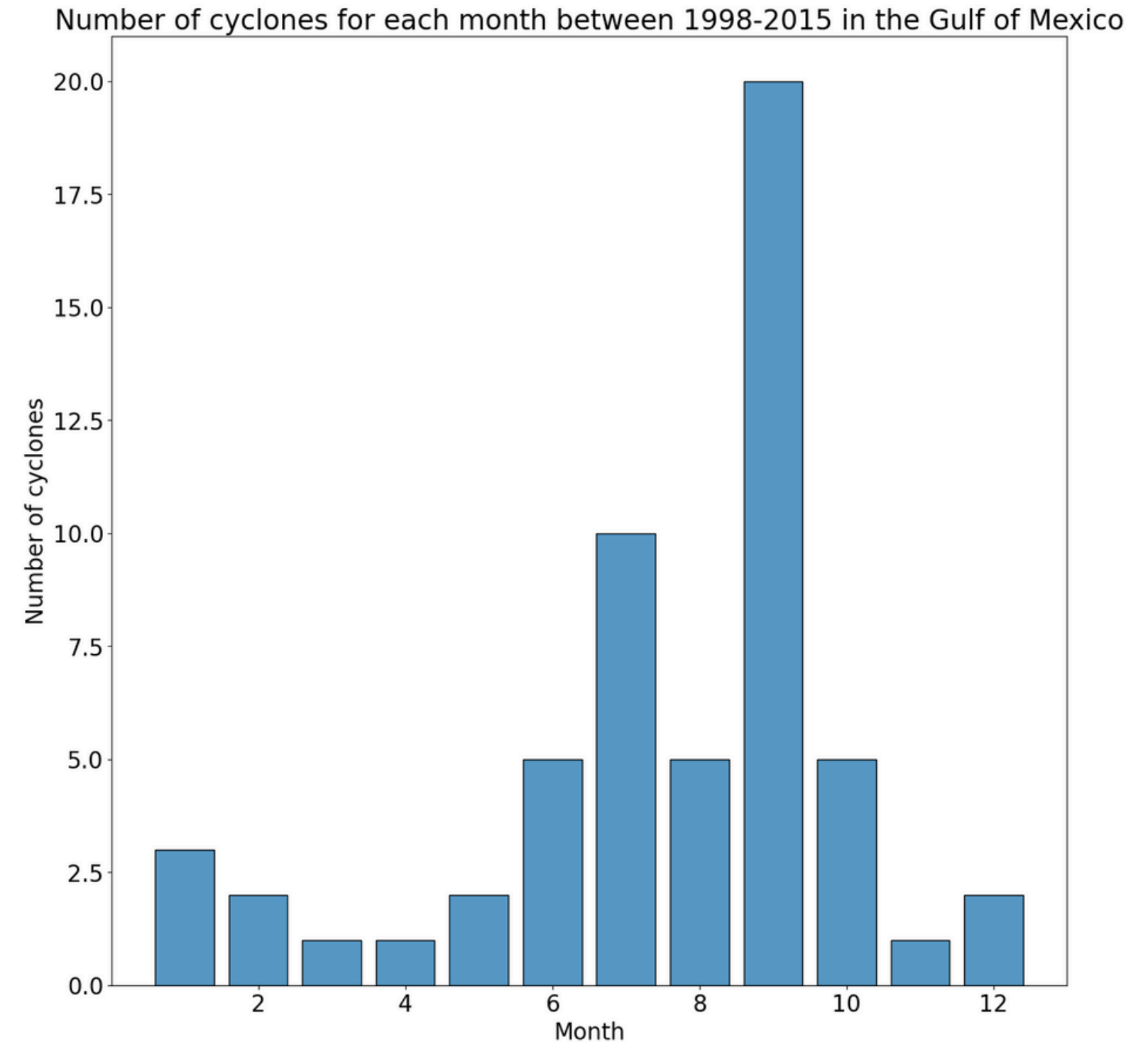
Source: *Historical Hurricane Tracks*

Second approach : Point by point

We can now calculate the number of cyclones for each month:



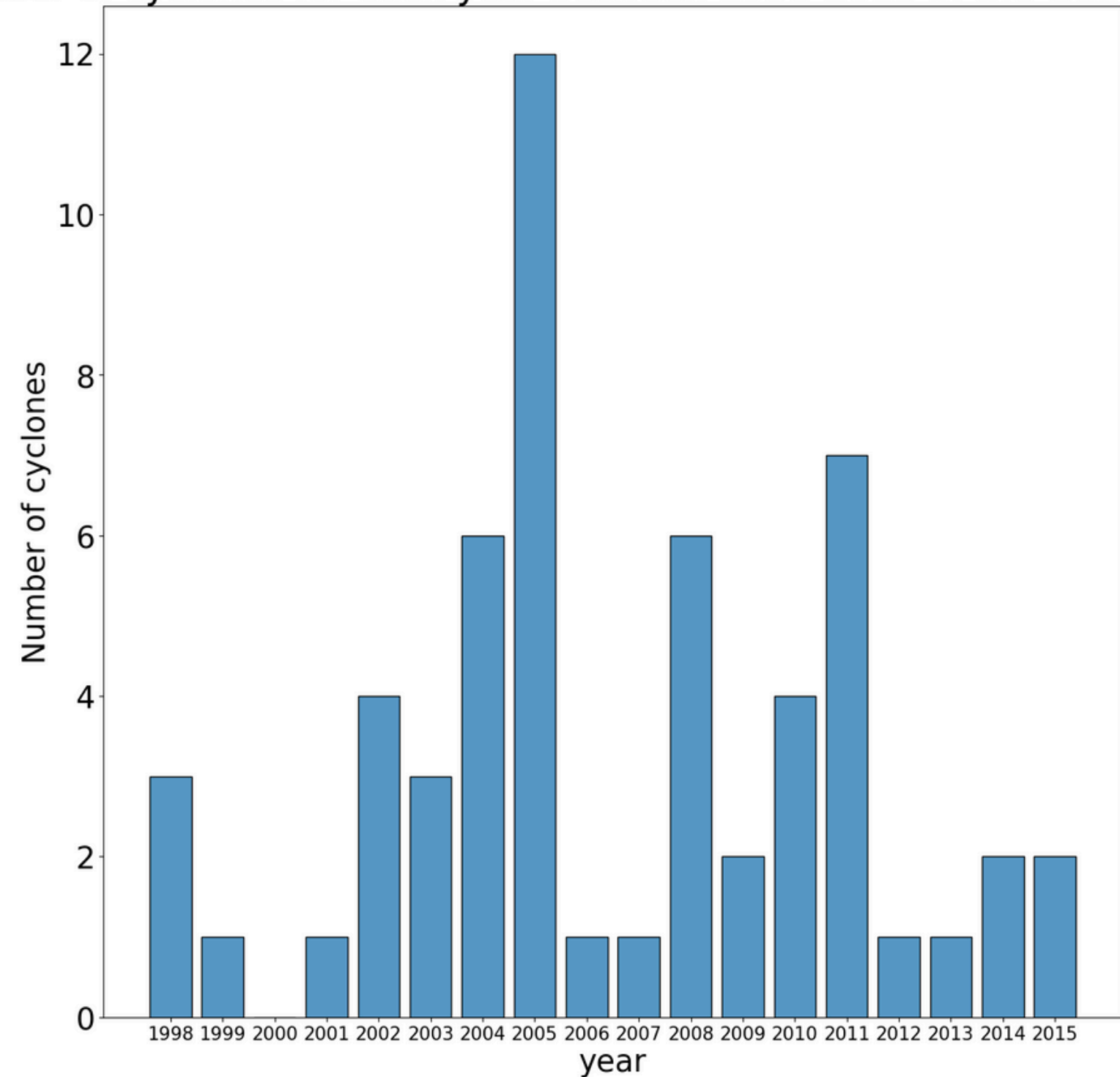
If at **least one cell signals a cyclone**, we count a cyclone



Second approach : Point by point

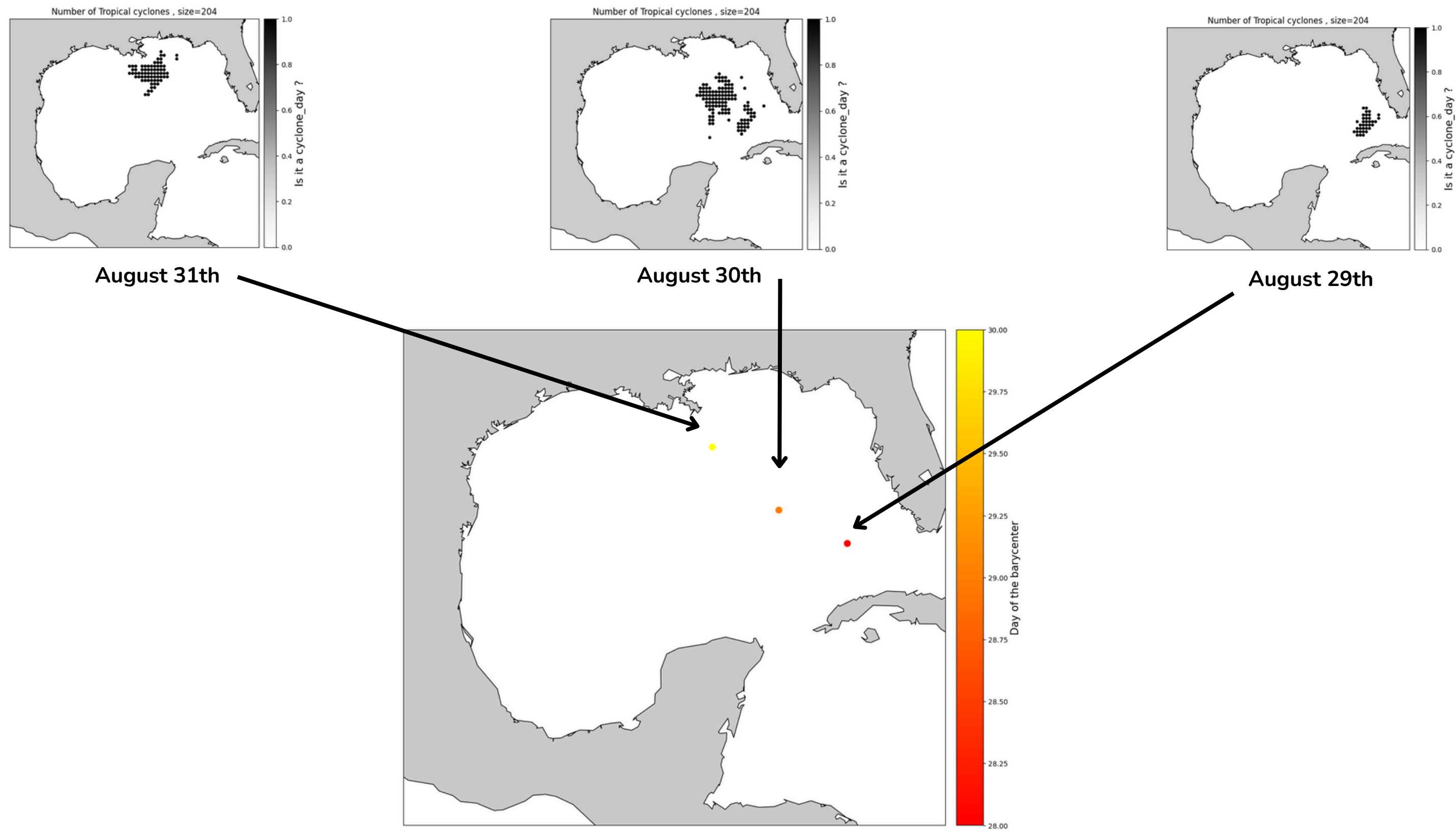
Similarly, we can calculate the number of cyclones per year

Number of cyclones for each year between 1998-2015 in the Gulf of Mexico



Seeing the Trajectory

Hurricane Katrina, August 29th, 30th, 31st 2005



Seeing the Trajectory

1

HURRICANE
EMILY 2005

Jul 11, 2005 to Jul 21, 2005

2

HURRICANE
KATRINA 2005

Aug 23, 2005 to Aug 31, 2005

3

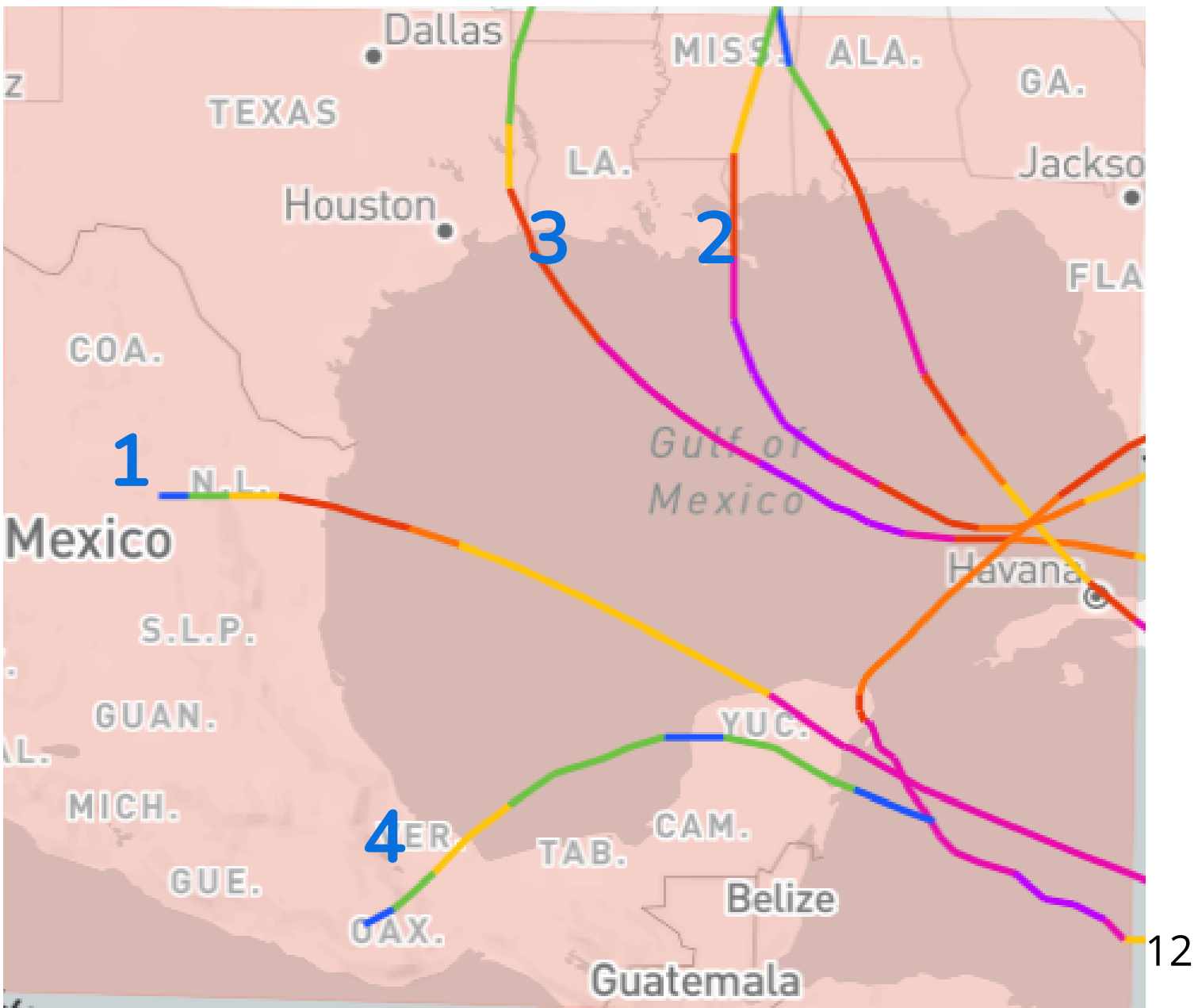
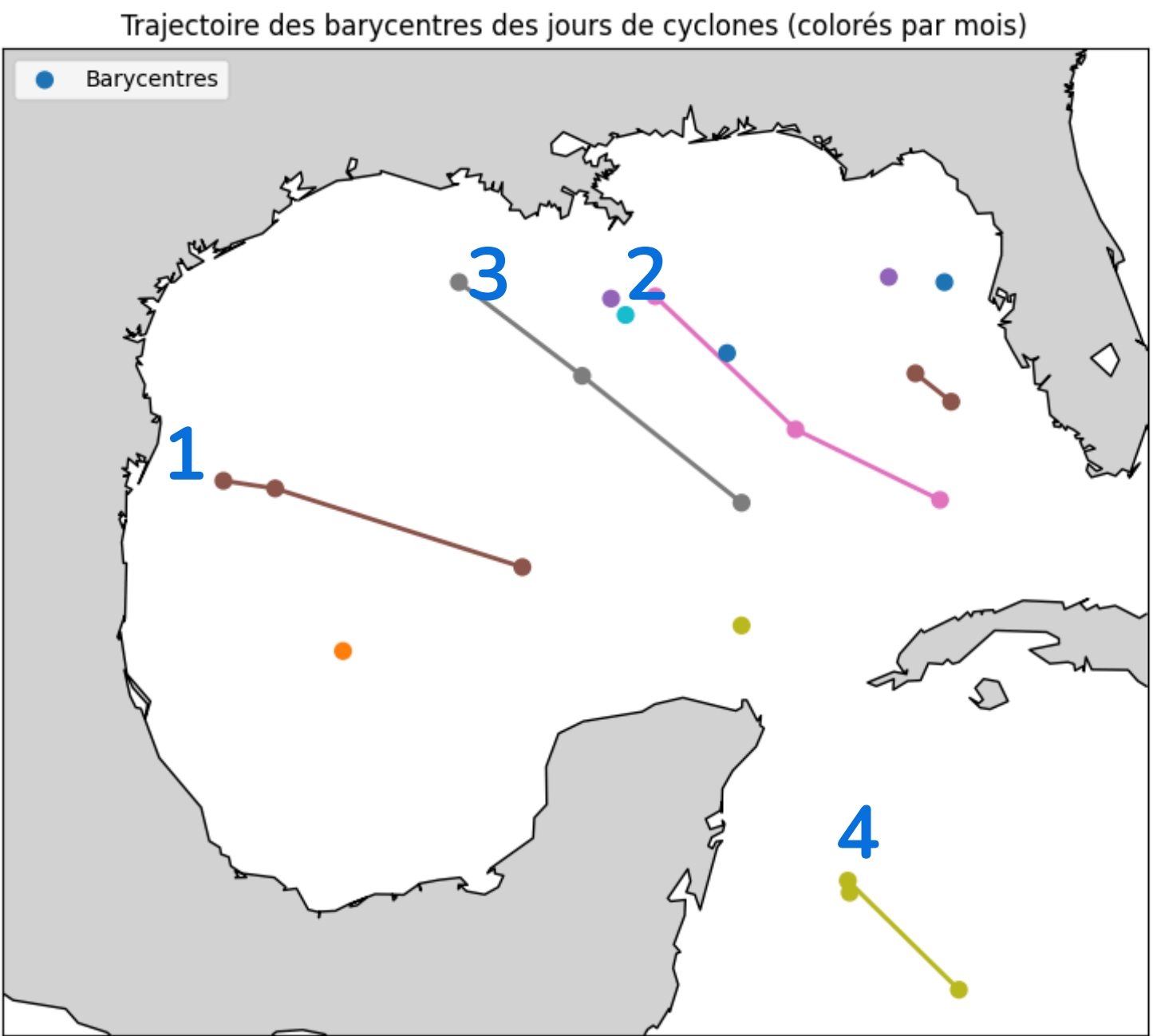
HURRICANE
RITA 2005

Sep 18, 2005 to Sep 26, 2005

4

HURRICANE
WILMA 2005

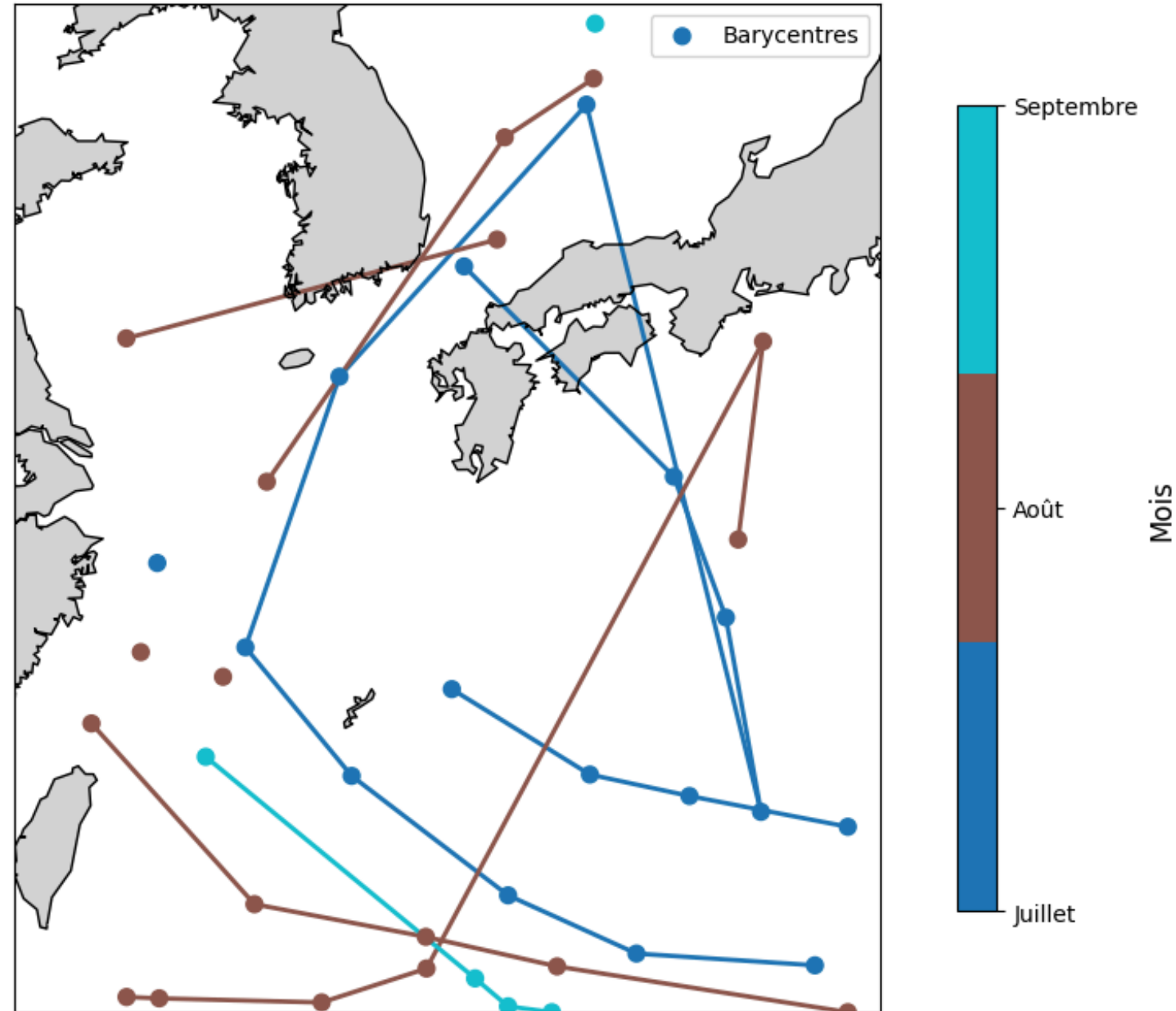
Oct 15, 2005 to Oct 26, 2005



Seeing the Trajectory

Japan Hurricanes: July, August, September 2015

Trajectoire des barycentres des jours de cyclones (Juillet, Août, Septembre)



Source: *Historical Hurricane Tracks*

Conclusion

- Using SST data, we can **detect tropical cyclones**
- **Following the temperature variations for each cell is the most effective methods**
- This methods also allows to **track the cyclone trajectory**

To go further...

- Given all the cyclone trajectory, developing a machine learning model for prediction
- Relevant to use a clustering method for hurricane identification
- Improve the result combining with the SST in situ data/physical modelling for missing points

